

AVINASH REDDY POLAM

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SUMMARY

Software Developer with 4+ years of experience engineering scalable **microservices** and **cloud-native backend architectures** using **Java (Spring Boot)**, **Apache Kafka**, and **PostgreSQL** in the **Finance** and **Technology** sectors. Skilled in integrating **LLMs** via **OpenAI APIs** and streamlining **CI/CD pipelines** using **Docker**, **Kubernetes**, and **Terraform** to automate workflows and reduce operational overhead.

EXPERIENCE

Software Developer | JPMorgan Chase & Co. | USA

August 2024 – Present

- Engineered production-grade microservices using **Java 17**, **Spring Boot**, and **PostgreSQL**, integrating **RESTful APIs** with **Prometheus** and **Grafana** to cut daily sync time by **19 hours** across **5 global platforms** while collaborating across time zones
- Streamlined **CI/CD automation** using **GitLab**, **Docker**, and **Terraform** with embedded scans and rollback logic to reduce release cycles by **3.5 hours** and support team alignment during retrospectives
- Designed an **LLM-powered parser** by integrating **OpenAI APIs**, **scikit-learn**, and **LangChain** into a **Java backend** to extract insights from **12,000+ documents** and reduce legal review time by **11 days per quarter**
- Led architecture reviews to define **monolith-to-Kubernetes migration strategy** and introduced **Helm** for reproducible deployments while mentoring **2 developers** and influencing infra planning across **2 core units**

Software Developer | Dell Technologies | India

May 2020 – July 2022

- Designed **Java-based Spring Boot microservices** for core orchestration workflows, scaling **backend services** to support 1.8 million daily active users.
- Structured persistent **data layers** and storage provisioning using **PostgreSQL**, reliably managing 3.5M+ telemetry and configuration records across environments.
- Implemented **event-driven API communication** via **Apache Kafka**, decoupling backend components to maintain **system stability** during high-demand feature launches.
- Automated **end-to-end testing frameworks** utilizing **Python** and **pytest** alongside **Java unit tests**, eliminating 8 hours of manual validation per release.
- Integrated **containerized workflows** using **Docker** and **CI/CD pipelines**, optimizing secure **service configurations** and standardizing deployments across staging and production.
- Coordinated technical milestones within cross-functional **Agile** pods, driving **code reviews** and **refactoring sessions** to enhance long-term **codebase maintainability**.

SKILLS

Programming Languages & Frameworks: Java (Java 17, Spring Boot, Spring Security, Hibernate, JPA, MVC, Microservices), Python, JavaScript, HTML5, CSS3, SQL, PL/SQL, JSON, XML, Shell Scripting

Frontend Development & UI/UX: React.js (JWT auth, REST integration), Thymeleaf, Responsive Web Design, Basic UI Debugging

Databases & Storage: PostgreSQL, Oracle DB, MySQL, SQL Server, JDBC

Cloud & DevOps: AWS (EC2, S3, Lambda), GitLab, Jenkins, Bitbucket, Docker, Terraform, Git, CI/CD Automation, Linux/Unix

AI, ML & Automation: OpenAI APIs, LangChain, LLM Integration, scikit-learn, TensorFlow (Java API), Weka, spaCy, Apache OpenNLP, Machine Learning Pipelines

Monitoring & Reliability: Prometheus, Grafana, Log Monitoring, Service Dashboards

APIs & Web Services: RESTful APIs, JWT, OAuth 2.0, Swagger/OpenAPI, Postman, Webhooks

Testing & Debugging: JUnit, Mockito, API Testing, Unit Testing, Integration Testing, Regression Testing

Performance & Security: Concurrency (ExecutorService, sync), Caching, Query Tuning, RBAC, Secure APIs

Integration & Middleware: Kafka, Keycloak, Azure Intune, API Integration, Modular Service Layers

Deployment & Infrastructure: Kubernetes, Helm, Maven, Gradle, Deployment Automation

Development Methodologies: Agile (Scrum), SDLC, Sprint Planning, Version Control, Peer Code Reviews, Stakeholder Collaboration

PROJECTS

Enhanced Security System with Breach Detection using Deep Learning

- Engineered a real-time **deep learning security system** using **TensorFlow**, **CNNs**, and **OpenCV** to accurately detect restricted-area breaches via multi-modal **biometric analysis**.

Student Analysis and Ranking Automation System

- Automated academic evaluations and **compliance reporting** by building a secure **Python** and **SQL**-based ranking platform with strict **audit trails**.

Data Augmentation Script – Triton Robotics

- Created a scalable **Python** and **Keras** utility that applies randomized **image transformations** to enrich and diversify datasets for **robotics computer vision models**.

EDUCATION

Texas A&M University - Commerce

Master of Science in Computer & Information Science

Commerce, TX

Aug 2022 – May 2024

CERTIFICATION

- Oracle Certified Associate, Java SE 8 Programmer
- AWS Certified Cloud Practitioner
- Forage – Walmart USA Advanced Software Engineering Job Simulation